Yujen Lan

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Senior Backend / Systems Software Engineer — specializing in Modern C++, Linux systems debugging, and large-scale codebase modernization, with quantified performance and reliability engineering results.

Summary

- Software Engineer with 3+ years of experience developing Linux-based system software and backend services using Modern C++ and Python. Strong expertise in C++, OOP, API design, concurrency, and Linux systems.
- Experienced in software architecture modernization, performance optimization, large-scale codebase maintenance, and system-level debugging.
- Identified structural blind spots in an industry-standard validation tool (DMTF Redfish Validator) — it couldn't verify field presence or operation ordering, only formatting — and designed a **new validation engine** to close the gap. Now covers 41 resource types and ~690 checks across real hardware, replacing ~40 manually-maintained modules.

Technical Skills

Software Engineering: Software Architecture | API Design | OOP | Design Patterns | RAII | Concurrency & Asynchronous Programming | Performance Optimization | System-Level Debugging | Refactoring
Platform Management Ecosystem: DMTF Redfish | OpenBMC | bmcweb | MCTP | PLDM | IPMI | DBus | Yocto / BitBake
Languages & Tools: C++, C, Python, Bash | CMake, Meson, CI/CD | Linux, Systemd | Git

Experience

Insyde Software | Software Engineer *July 2024 – Present*

- **Reduced platform startup time by over 60%** through startup profiling, dependency analysis, and service initialization optimization.
- **Designed a scalable integration workflow** with an upstream open-source platform that reduced feature migration effort from approximately month to week while eliminating long-term patch maintenance overhead.
- **Modernized legacy C services into modular Modern C++ architectures**, reducing technical debt and improving longterm maintainability.
- **Refactored API schemas and data models**, improving maintainability, readability, and runtime performance by approximately 30%.
- **Built reusable utility libraries** for resource management, file operations, and lifecycle control using RAII, OOP design patterns, and modern C++ practices.
- **Resolved production stability issues** including service crashes, object lifetime defects, resource ownership problems, and concurrency-related issues.
- **Improved standards compliance and interoperability** by resolving API validation and compatibility issues.
- **Reorganized multi-repository projects and build configurations** while preserving development history (Git) and improving maintainability.
- **Upgraded legacy back-end service architecture** to a lightweight reverse-proxy model, improving deployment flexibility and operational simplicity.
- **Simplified deployment workflows** through unified API-driven operations, reducing operational complexity and improving user experience.

- **Lead development of an automated validation platform** capable of validating complex API workflows beyond existing industry validation tools.
  GitHub: https://github.com/JohnBlue-git/RedfishAutoYamlTestInterview
- Diagnose and resolve cross-layer issues spanning applications, middleware, operating systems, networking, and platform infrastructure.
- Collaborate with cross-functional engineering teams to deliver new features, improve software quality, and reduce technical debt.
- Author technical documentation, API specifications, and developer guides for internal teams and external partners.

SinoPac Securities | Software Engineer *June 2023 – April 2024*

- Developed and maintained backend applications using C++ on Linux platforms.
- **Improved data-processing efficiency by 3 to 4 times** through algorithm optimization and workflow redesign.
- Built automated CI/CD pipelines integrating build validation and testing processes, reducing manual effort and improving software quality.

Education

National Cheng Kung University (NCKU)

Master Degree *in Systems and Naval Mechatronic Engineering (Year: 2021 GPA: 4.3)*

Bachelor Degree *in Systems and Naval Mechatronic Engineering (Year: 2018 GPA: 3.7)*

Open Source Projects

MultiFish

Experimental project exploring REST API architecture, backend service design, and modern Golang system development.
Github: https://github.com/JohnBlue-git/MultiFish

HowToSDBusPlus

Practical guide and implementation examples for modern C++ DBus service development, inter-process communication, and Linux system programming.
GitHub: https://github.com/JohnBlue-git/HowToSDBusPlus

GoVarlink

Implementation examples and experiments focused on lightweight IPC mechanisms and service communication.
GitHub: https://github.com/JohnBlue-git/GoVarlink

HowToYocto

Documentation and practical examples covering Linux build systems, software integration, and platform customization.
GitHub: https://github.com/JohnBlue-git/HowToYocto

HowToPLDM

Technical notes and implementation examples focused on protocol architecture and system integration concepts.
GitHub: https://github.com/JohnBlue-git/HowToPLDM